Chapter 10

Inventories

Introduction

In general terms, inventories are stocks of goods. For a manufacturing business, these will usually consist of raw materials, partly-completed products (work in progress) and finished products awaiting sale to customers. For a retailer or wholesaler, inventories will comprise mainly stocks of goods acquired for resale.

The value placed on inventories affects the cost of sales figure shown in the statement of comprehensive income and so has a direct impact on reported profit. Furthermore, the value of inventories is often very significant, so that even a small percentage error in their measurement could lead to a material mis-statement of profit. IAS2 *Inventories* provides guidance on the valuation of inventories and the purpose of this short chapter is to explain the main requirements of that standard.

Contracts with customers

IFRS15 *Revenue from Contracts with Customers* prescribes the accounting treatment of customer contracts. This standard identifies the circumstances in which a sale of goods is deemed to have occurred, so that sales revenue is recognised and the associated inventories are derecognised. The requirements of IFRS15 are explained in Chapter 13 of this book.

Objectives

By the end of this chapter, the reader should be able to:

- define the term "inventories" and determine the cost of inventories in accordance with the requirements of IAS2 *Inventories*
- apply cost formulas such as first-in, first-out and weighted average cost
- determine the net realisable value of inventories and correctly measure inventories at the lower of cost and net realisable value
- list the main disclosures relating to inventories that should be made in the notes to the financial statements.

Inventories

IAS2 defines inventories as "*assets:*

(a) held for sale in the ordinary course of business;

(b) in the process of production for such sale; or

(c) in the form of materials or supplies to be consumed in the production process or in the rendering of services".

The standard also prescribes a well-known rule for the valuation of inventories. This rule is that "*inventories shall be measured at the lower of cost and net realisable value*". The remainder of IAS2 provides guidance on the determination of the cost and net realisable value of inventories and makes certain disclosure requirements.

Cost of inventories

IAS2 states that the cost of inventories "*shall comprise all costs of purchase, costs of conversion and other costs incurred in bringing the inventories to their present location and condition*". Note the following points:

(a) Costs of purchase consist of the purchase price of inventories, including import duties and other taxes (if these are irrecoverable), transport and handling costs and any other costs directly attributable to the acquisition of the goods concerned. Trade discounts are deducted when calculating costs of purchase.

(b) Costs of conversion are relevant mainly to manufacturing businesses and consist of direct costs of production (such as direct labour) together with a systematic allocation of fixed and variable overheads incurred in converting materials into finished goods.

Fixed production overheads are indirect production costs that remain relatively constant regardless of the volume of production (e.g. factory rent and rates). Variable production overheads are indirect production costs that vary more or less in line with the volume of production, such as indirect materials and some indirect labour costs.

(c) The allocation of fixed production costs to units of production should be based upon the normal capacity of the production facilities.

(i) If production is abnormally low, there will be unallocated fixed production overheads. These should be recognised as an expense in the period in which they are incurred and should *not* be included in the cost of inventories.

(ii) If production is abnormally high, the amount of fixed production overheads allocated to each item of production should be decreased so as to ensure that the cost of inventories is not overstated.

(d) Certain costs are explicitly excluded from the cost of inventories. The main items are:

(i) the costs of abnormal wastage of materials, labour or other production costs

(ii) storage costs, unless these costs are necessary in the production process before a further production stage

(iii) administrative overheads that do not contribute to bringing inventories to their present location or condition

(iv) selling costs.

(e) Techniques such as standard costing and the retail method may be used to determine the cost of inventories, so long as they provide a reasonable approximation to cost.

(i) Standard costs are computed in accordance with normal prices and usage of materials, labour etc. and normal levels of efficiency. Standard costs should be reviewed regularly and revised if necessary.

(ii) The retail method is often used in the retail industry for measuring inventories which consist of large numbers of rapidly changing items with similar profit margins. This method determines the cost of inventories by reducing their sales value by the appropriate percentage profit margin.

EXAMPLE 1

A manufacturing company's inventory at the end of an accounting period includes a finished product with the following costs to date:

	£
Purchase price of materials	9,000
Less: 5% trade discount	450
	8,550
Import duty at 10%	855
Direct labour costs	5,850
Allocation of fixed production overheads	2,840
Storage costs since product completed	225
Advertising costs	317
Total cost	18,637

Notes:

(i) All of these costs exclude VAT. The company is able to recover any VAT which it is charged by its suppliers but cannot recover import duties.

(ii) One-third of the materials included above were wasted as the result of an abnormal machine malfunction.

(iii) Fixed production overheads are allocated to products on the assumption that production is running at normal capacity. In fact, production was unusually low during the period in which this product was made and a further allocation of £820 would be required if fixed overheads were allocated on the basis of actual production.

(iv) The advertising costs were incurred whilst trying to find a buyer for the product. No buyer had been found by the end of the accounting period.

Calculate the cost of this product in accordance with the requirements of IAS2.

Solution

For inventory valuation purposes, the cost of the product is as follows:

	£
Purchase price of materials, net of trade discount	5,700
Import duty at 10%	570
Direct labour costs	5,850
Allocation of fixed production overheads	2,840
Total cost	14,960

Notes:

(i) Recoverable VAT is not a cost to the company and is therefore excluded from the cost of inventories.

(ii) The cost of abnormal wastage is also excluded from the cost of inventories. The cost of the remaining two-thirds of the materials was £5,700 (2/3 x £8,550).

(iii) The further £820 of fixed production overheads is treated as an expense and is not included in the cost of inventories.

(iv) The storage costs were incurred after production was complete. The advertising costs are selling costs. Neither of these can be included in the cost of inventories. Furthermore, the failure to find a buyer for the product may cast doubt on its net realisable value (see later in this chapter).

Cost formulas

In general, IAS2 requires that the cost of inventory items should be ascertained "*by using specific identification of their individual costs*". In other words, each item of inventory should be considered separately and its cost should be individually established.

However, it is not always possible to distinguish one inventory item from another. For instance, a business may have an inventory of identical machine parts which have been bought on different dates and at different prices. In such a case, it is frequently impossible to determine the cost of each individual item with any certainty. IAS2 refers to such items as "interchangeable" and requires that their cost should be ascertained by means of a cost formula. Two cost formulas are allowed. These are:

(a) **First-in, first-out (FIFO)**. The FIFO formula assumes that inventory items which are purchased or produced first are sold or used first, so that the items remaining at the end of an accounting period are those most recently purchased or produced.

(b) **Weighted average cost (AVCO)**. Use of the AVCO formula generally involves computing a new weighted average cost per item after each acquisition takes place. The cost of the items remaining at the end of an accounting period is then calculated by using the most recently-computed weighted average cost per item.

IAS2 allows either of these cost formulas to be used and there is no requirement that the choice of formula should mirror the actual physical flow of the items concerned. However, there is a requirement that the same cost formula should be used for all inventories having a similar nature and use.

LIFO not allowed

It is important to note that IAS2 does *not* allow the use of last-in, first-out (LIFO). This cost formula assumes that the newest inventory items are sold or used first, so that the items remaining at the end of an accounting period are the oldest. IAS2 objects to the LIFO formula for a number of reasons:

(a) LIFO is generally not a reliable representation of actual inventory flows.

(b) The use of LIFO is often tax-driven, since it matches sales revenue against the cost of the newest items acquired. In times of rising prices, this will result in lower reported profits and a lower tax liability (so long as the tax authorities will accept LIFO for tax purposes). However, the IASB takes the view that tax considerations do not provide an adequate conceptual basis for the selection of an accounting treatment.

(c) The use of LIFO results in inventories being shown in the financial statements at values which bear little relation to recent cost levels.

Even though the use of LIFO has been eliminated, IAS2 does not rule out "*specific cost methods that reflect inventory flows that are similar to LIFO*". For instance, inventories may be held in piles or bins which are both depleted and replenished from the top. In such a case, it could be argued that there is no need to apply a cost formula at all, since the items remaining on the pile or in the bin can be clearly identified. A specific costing approach which reflected the physical flow of items in these circumstances would not be ruled out purely because it gave similar results to LIFO.

EXAMPLE 2

On 1 April 2019, a company's inventory included 12,000 bricks which had been acquired for £140 per thousand. Purchases and sales of bricks during the year to 31 March 2020 were as follows:

Purchases	*Number bought*	*Cost per thousand*	*Sales*	*Number sold*
		£		
14 July 2019	10,000	142	26 August 2019	16,000
11 October 2019	8,000	145	5 November 2019	7,000
19 January 2020	15,000	147	12 March 2020	17,000

Calculate the cost of the bricks sold during the year and the cost of the inventory of bricks remaining at 31 March 2020, using:

(a) first-in, first-out (FIFO)

(b) last-in, first-out (LIFO)

(c) weighted average cost (AVCO).

Which of these three cost formulas should the company use?

Solution

(a) **FIFO**

	Number of bricks			*Cost (£)*
Sold 26 August	16,000	12,000 @ 14.0p	1,680	
		4,000 @ 14.2p	568	2,248
Sold 5 November	7,000	6,000 @ 14.2p	852	
		1,000 @ 14.5p	145	997
Sold 12 March	17,000	7,000 @ 14.5p	1,015	
		10,000 @ 14.7p	1,470	2,485
Cost of goods sold				5,730
Inventory at 31 March	5,000	5,000 @ 14.7p		**735**

(b) **LIFO**

	Number of bricks			*Cost (£)*
Sold 26 August	16,000	10,000 @ 14.2p	1,420	
		6,000 @ 14.0p	840	2,260
Sold 5 November	7,000	7,000 @ 14.5p		1,015
Sold 12 March	17,000	15,000 @ 14.7p	2,205	
		1,000 @ 14.5p	145	
		1,000 @ 14.0p	140	2,490
Cost of goods sold				5,765
Inventory at 31 March	5,000	5,000 @ 14.0p		**700**

(c) **AVCO**

	Number of Bricks		Total cost	Weighted average	Cost (£)
Opening inventory	12,000	@ 14.0p	1,680		
Bought 14 July	10,000	@ 14.2p	1,420		
	22,000		3,100	14.09p	
Sold 26 August	16,000	@ 14.09p	2,254		2,254
	6,000		846		
Bought 11 October	8,000	@ 14.5p	1,160		
	14,000		2,006	14.33p	
Sold 5 November	7,000	@ 14.33p	1,003		1,003
	7,000		1,003		
Bought 19 January	15,000	@ 14.7p	2,205		
	22,000		3,208	14.58p	
Sold 12 March	17,000	@ 14.58p	2,479		2,479
Cost of goods sold					5,736
Inventory at 31 March	5,000	@ 14.58p	729		**729**

The company may use either FIFO or AVCO but must use the same cost formula for all inventories of a similar nature and use. The business is not allowed to use LIFO.

Net realisable value

As stated above, IAS2 requires that inventories should be measured at the lower of cost and net realisable value. Net realisable value (NRV) is defined as "*estimated selling price in the ordinary course of business less the estimated costs of completion and the estimated costs necessary to make the sale*". The effect of measuring inventories at the lower of cost and NRV is that any loss expected on the sale of items at less than cost is accounted for immediately rather than when the items are sold. Note that:

(a) The costs and NRV of inventories are normally compared item by item. However, IAS2 accepts that it may sometimes be appropriate to group similar or related items together and to compare the total cost of the group with its total NRV.

(b) Materials and other supplies held for use in production are not written down below cost so long as the finished products in which they will be incorporated are expected to be sold at cost or above.

EXAMPLE 3

The inventory of a motor vehicles dealer at the end of an accounting period includes the following used vehicles:

	Costs incurred to date	*Expected further costs before sale*	*Expected selling price*
	£	£	£
Vehicle A	14,200	1,250	18,000
Vehicle B	17,500	1,000	20,000
Vehicle C	11,900	1,240	14,000
Vehicle D	13,000	2,760	15,000

The dealer's sales staff are paid a commission when they sell a vehicle. This commission is calculated at 5% of selling price. Calculate the value at which these items should be shown in the dealer's financial statements.

Solution

	Cost	*NRV*	*Lower of cost and NRV*
	£	£	£
Vehicle A	14,200	(18,000 x 95%) – 1,250 = 15,850	14,200
Vehicle B	17,500	(20,000 x 95%) – 1,000 = 18,000	17,500
Vehicle C	11,900	(14,000 x 95%) – 1,240 = 12,060	11,900
Vehicle D	13,000	(15,000 x 95%) – 2,760 = 11,490	11,490
	56,600	57,400	55,090

For vehicles A, B and C, cost is less than NRV, so these vehicles are valued at cost. But the cost of vehicle D exceeds its NRV, so this vehicle must be valued at NRV. The closing inventory should be valued at a total of £55,090. It would not be permissible to value the vehicles at the lower of total cost and total NRV (i.e. £56,600).

Disclosures relating to inventories

The main disclosures required by IAS2 are as follows:

(a) the accounting policies adopted in measuring inventories, including cost formulas

(b) the total carrying amount of inventories (i.e. the amount at which the inventories are shown or "carried" in the entity's financial statements) together with an analysis into appropriate categories

(c) the amount of inventories recognised as an expense during the period

(d) the amount of any write-down to net realisable value during the period and the amount of any reversal of previous write-downs.

These disclosures will be made in the notes which form part of the financial statements.

Summary

- International standard IAS2 defines "inventories" and requires that inventories are measured at the lower of cost and net realisable value.
- The cost of inventories comprises costs of purchase, conversion costs and any other costs incurred in bringing the inventories to their present location and condition.
- The cost of interchangeable inventory items should be established by use of a cost formula. Permitted formulas are first-in, first-out and weighted average cost. The use of last-in, first-out is not permitted.
- The net realisable value of inventories is their estimated selling price in the ordinary course of business, less estimated costs of completion and selling costs.
- IAS2 makes certain disclosure requirements with regard to inventories.

Exercises

A set of multiple choice questions for this chapter is available on the accompanying website.

10.1 (a) Explain the term "inventories" as defined by international standard IAS2.

(b) List the costs which should be included when measuring the cost of inventories and identify any costs which should be excluded.

(c) Explain the term "net realisable value" in relation to inventories.

10.2 (a) Identify the circumstances in which a cost formula may be used to establish the cost of inventories.

(b) A company's inventories at 30 April 2020 include 11,000kg of a chemical which is used in the company's manufacturing processes. Purchases and issues of this chemical during the year to 30 April 2020 were as follows:

		Number of kg	*Cost per kg*
May 2019	Opening inventory	32,000	£7.50
June 2019	Issued to production	25,000	
August 2019	Purchased	15,000	£8.75
September 2019	Issued to production	12,000	
October 2019	Issued to production	5,000	
January 2020	Purchased	10,000	£9.50
February 2020	Issued to production	4,000	

Calculate the cost of the company's inventory of the chemical at 30 April 2020 using each of the following cost formulas:

(i) first-in, first-out (FIFO)

(ii) weighted average cost (AVCO).

10.3 Kellerstone Ltd buys used machines which it reconditions and sells on to customers. The company's inventory at the end of its most recent accounting period included the following machines:

	Purchase price	*Reconditioning costs incurred to date*	*Expected further costs before sale*	*Expected selling price*
	£	£	£	£
Machine W	10,300	650	-	12,500
Machine X	15,750	850	400	17,500
Machine Y	18,450	500	2,100	23,000
Machine Z	8,300	-	900	9,500
	52,800	2,000	3,400	62,500

Selling expenses are expected to absorb 4% of the expected selling price.

(a) Explain why it is not permissible to value the inventory of machines at the lower of their total purchase price (£52,800) and their total net realisable value.

(b) Compute the value at which the inventory of machines should be measured in the company's financial statements.

10.4 A company which makes only one type of product incurs fixed production overheads of £240,000 during the year to 31 March 2020. Normal production capacity is 80,000 units per annum. Calculate the amount of fixed production overheads that should be allocated to each unit of production when determining the cost of inventories at 31 March 2020 if actual production for the year is:

(a) 80,000 units (b) 100,000 units (c) 50,000 units.

*__10.5__ Parrison Ltd is a manufacturing company. The company's inventory at 30 June 2020 includes the following items of work in progress:

	Product X	*Product Y*
Number of units held at 30 June 2020	6,400	3,800
Costs incurred per unit to date	£8	£12
Estimated further costs per unit to completion	£7	£8
Estimated selling costs per unit	£2	£4
Estimated selling price per unit	£15	£30

The company complies with IAS2 *Inventories*.

(a) Define the term "inventories".

(b) Explain how the cost of inventories should be determined.

(c) Define the term "net realisable value".

(d) Explain how inventories should be measured in accordance with IAS2.

(e) Calculate the figure at which the work in progress of Parrison Ltd should be shown in the company's financial statements for the year to 30 June 2020.

***10.6** The main activity of J&T Ltd is to buy old vehicles which are sold after converting them into a saleable condition. At 29 February 2020 (the end of the company's financial year) J&T Ltd had the following vehicles that were at various stages in the process of being converted to be ready for sale.

Vehicle	*Purchase price*	*Costs incurred on bringing to present location*	*Conversion costs incurred to date*	*Expected further costs before sale*	*Expected selling price*
	£	£	£	£	£
W	9,470	1,080	880	-	14,000
X	12,830	940	1,540	150	19,300
Y	3,550	750	1,260	600	6,000
Z	7,680	460	-	2,200	11,800

Notes:

1. Expected selling expenses for each vehicle are 6% of the expected selling price.
2. A quarter of the conversion costs relate to materials. But 20% of the cost of the materials used in the conversions is considered to be abnormal wastage, due to a poor quality type of material that has been used.

In an attempt to reduce the abnormal wastage of materials, material P has been used by the company since 5 January 2020, when 25 kilos were purchased for a total cost of £550. Subsequent purchases and usage of this material were as follows:

Date	*Transaction*	*Number of kilos*	*Total cost (£)*
Jan 7	Used	19	
Jan 20	Purchased	20	520
Jan 28	Used	18	
Feb 7	Used	4	
Feb 15	Purchased	15	315
Feb 29	Used	3	

To ensure that the correct valuation of the company's inventories is reflected in the final accounts, the company follows the guidance provided in IAS2 *Inventories*.

Required:

(a) Compute, in accordance with IAS2 *Inventories*, the value at which the inventory of vehicles as at 29 February 2020 should be shown in the final accounts of J&T Ltd.

(b) Determine the total cost of the abnormal wastage of materials that has been incurred on the conversion of the four vehicles.

(c) Calculate the total cost of material P that has been used and the cost of the inventory of this material as at 29 February 2020, using each of the following cost formulas:

(i) First-in, first-out (FIFO) (ii) Weighted average cost (AVCO).

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